Brian Connor Liau

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Education

Tufts University, School of Engineering Medford, MA

Bachelor of Science in Electrical Engineering, Minor in Embedded Systems Sept. 2023 – May 2027

GPA: 3.91/4.0 — **Dean's List:** Fall 2023 – Spring 2026

Coursework: Embedded Systems, Digital Signal Processing, Digital IC Design, Intro to VLSI, Electronics I & II

Technical Skills

- **Hardware & Lab Tools:** Oscilloscopes, Multimeters (DMM), Function Generators, Power Supplies, Logic Analyzers, SMUs, Surface Mount Soldering/Rework
- **PCB & Simulation Tools:** Altium Designer, KiCad, LTspice, HSPICE, Schematic Capture, PCB Layout
- **Protocols & Hardware:** I2C, SPI, UART, BLE / 2.4 GHz RF, ESP32, STM32, Raspberry Pi, Arduino
- **Languages & Software:** C, C++, Python, MATLAB, SystemVerilog, VHDL, Git, Zephyr RTOS

Engineering Experience & Research

Silicon XPandas Taipei, Taiwan

Electrical Engineering Intern May 2025 – Aug. 2025; June 2026 – Aug. 2026

- Validated multi-transceiver hardware systems using oscilloscopes and RF test tools; analyzed test data to resolve signal integrity failures and improve test reliability.
- Executed board bring-up and developed low-level C validation scripts to verify timing, protocol compliance, and signal integrity for I3C interfaces.
- Developed Python-based GUI test automation scripts and shell runners, processing test logs to analyze IC chip performance and improving verification efficiency by 40%.

Center for Interactive Technologies (CITI), Univ. of São Paulo São Paulo, Brazil

Summer Research Intern May 2026 – June 2026

- Developed a custom Zephyr RTOS driver for an on-chip SHA-256 hardware accelerator on Puppy v2.
- Refactored legacy bare-metal driver code into a thread-safe Zephyr crypto API managing CSRs and interrupts.
- Validated hardware-accelerated hashing performance against software libraries, reducing CPU execution overhead.

Hardware & Embedded Systems Projects

Axis Guitar Project (IDEA Lab) Sept. 2025 – May 2026

Embedded Hardware Engineer *Altium Designer, KiCad, LTspice, ESP32, I2C, Lab Equipment*

- Executed schematic capture and 2-layer PCB layout using KiCad, integrating an ESP32 microcontroller, IMU sensor, and power management circuits.
- Modeled power supply stability and analog front-end signals in LTspice prior to physical fabrication.
- Led hands-on board bring-up, debugging, and fine-pitch SMT component rework; used oscilloscopes and multimeters to analyze signal integrity and resolve I2C bus timing issues, cutting latency by 35%.

Pipelined RV32I Processor Core May 2026 – Present

RTL Designer *SystemVerilog, OpenLane 2, Sky130*

- Microarchitected a 5-stage pipelined RISC-V (RV32I) core in SystemVerilog with hazard detection, data forwarding, pipeline stalls, and branch flushing.
- Integrated separate instruction and data memory interfaces mapped onto dedicated AHB-Lite Manager ports.
- Optimized total core power to 1.25 mW @ 95.2 MHz; validated system using SystemVerilog Assertion (SVA).

AMBA AHB-Lite Bus Interconnect & Subordinate IP July 2026 – Aug. 2026

RTL Designer *SystemVerilog, Verilator, OpenLane 2*

- Designed a reusable, parameterized AMBA AHB-Lite interconnect supporting dual-port SRAM memory subordinates with dynamic wait-state generation and multi-size access decoding.
- Authored formal SVA checkers enforcing ARM AHB-Lite protocol compliance, validating control/address/data stability during wait-states.